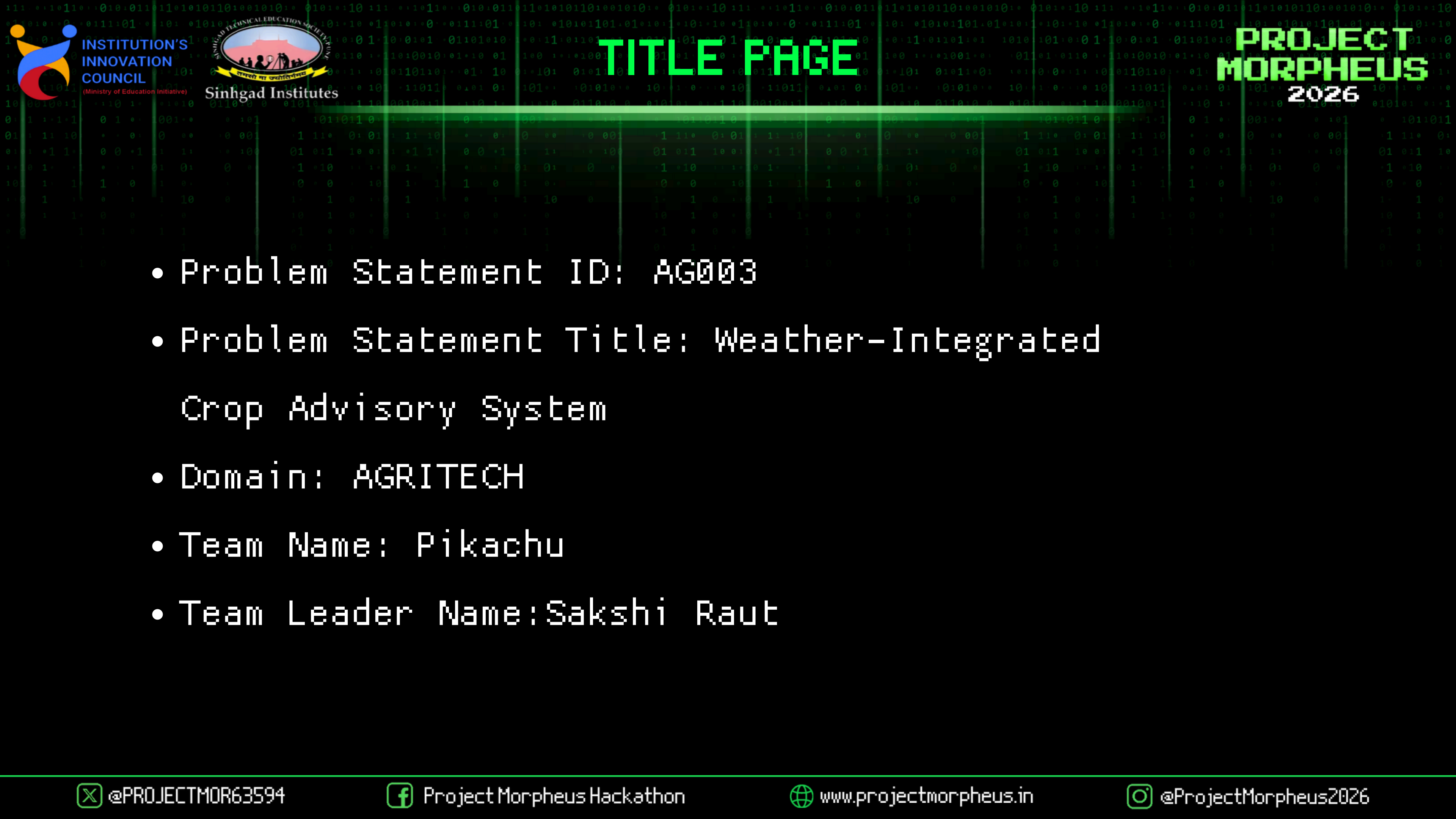




INSTITUTION'S
INNOVATION
COUNCIL
(Ministry of Education Initiative)



TITLE PAGE

PROJECT
MORPHEUS
2026

- Problem Statement ID: AG003
- Problem Statement Title: Weather-Integrated Crop Advisory System
- Domain: AGRITECH
- Team Name: Pikachu
- Team Leader Name: Sakshi Raut

PROPOSED SOLUTION: KRISHI MITRAM

Krishi Mitryam is an AI-driven Weather-Integrated advisory platform that works across App, WhatsApp, SMS, and AI Agentic Caller for keypad phones, making it inclusive for all farmers.

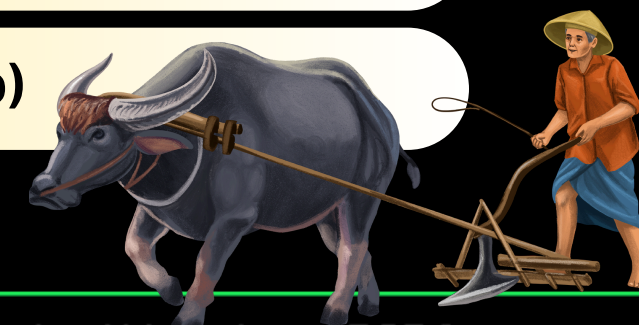
- Multi-channel Access
- Multilingual Support
- Image & Voice Inputs
- RAG-Based Verified Advisory Engine
- Context Awareness & Escalation System
- Weather-Integrated

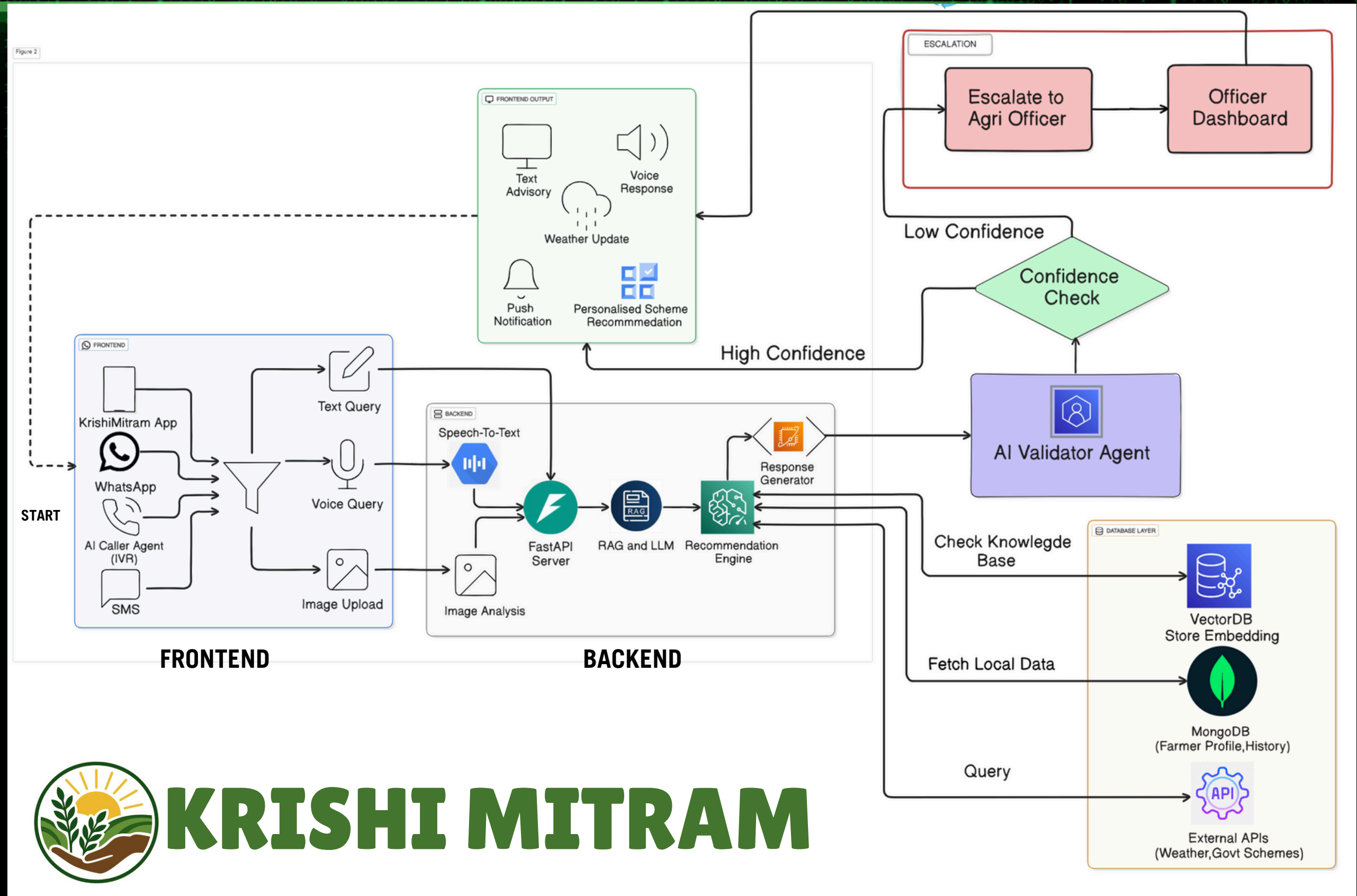
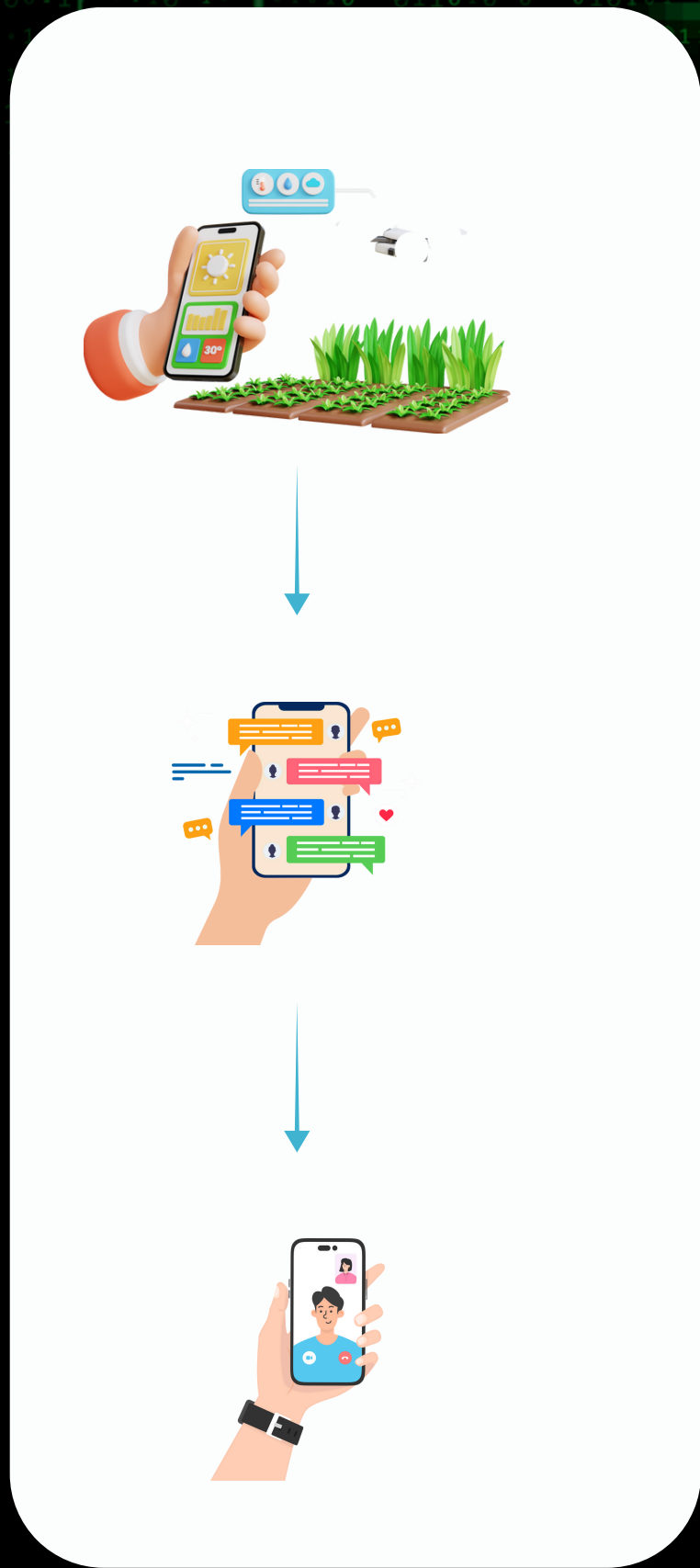
Krishi Mitryam is a Digital Climate Krishi Officer for India — an AI-powered, weather-integrated advisory system accessible through App, WhatsApp, SMS, and AI Voice Calls (for both smartphone and keypad users). Farmers can ask queries in their local language using voice, text, or images and receive instant, personalized guidance on crop practices, pest control, weather risks, and government schemes.

It uses a Hybrid AI approach combining Real-Time Weather APIs, RAG-based verified retrieval, an AI Validator Agent, and rule-based agronomy to ensure accurate, safe, and climate-smart recommendations. With offline-first design and multi-channel delivery, Krishi Mitryam is highly feasible and ensures no farmer is left behind in the journey toward climate-resilient smart agriculture.

INNOVATION & UNIQUENESS:

- Multilingual-first AI system, scalable to Indian languages
- Hybrid AI model → LLM + Computer Vision + Rule-based agronomy
- Offline-first design for low connectivity areas
- Website, Mobile App, WhatsApp, SMS, AI Voice Calls
- AI + Human collaboration (escalation & feedback loop)







METHODOLOGY USED

PROJECT
MORPHEUS
2026

1

Multi-Channel Delivery Layer

- Mobile App + Website
- WhatsApp & SMS chatbot
- AI Voice Caller for keypad phones
- Ensures accessibility for all literacy levels.

2

Multilingual NLP Pipeline

- Speech/Text queries supported in Indian languages
- Language detection + translation layer
- Fine-tuned LLM for agriculture domain responses
- Multilingual-first advisory system.

3

Hybrid AI Intelligence Model

Krishi Mitryam uses:

- LLM-based reasoning (crop advice, schemes, Q&A)
- Computer Vision (leaf disease detection from images)
- Rule-Based Agronomy Engine (fertilizer limits, safe pesticide dosage)

4

Retrieval-Augmented Generation (RAG)

Knowledge Engine

- Government advisories, ICAR research, weather data, market rates
- Vector database retrieval + LLM response generation
- Answers are grounded in verified sources.

4

Weather & Context-Aware Decision Support

- Real-time integration with IMD weather APIs
- Location-based crop recommendations
- Pest outbreak alerts using climate patterns
- Personalized and climate-smart guidance.

5

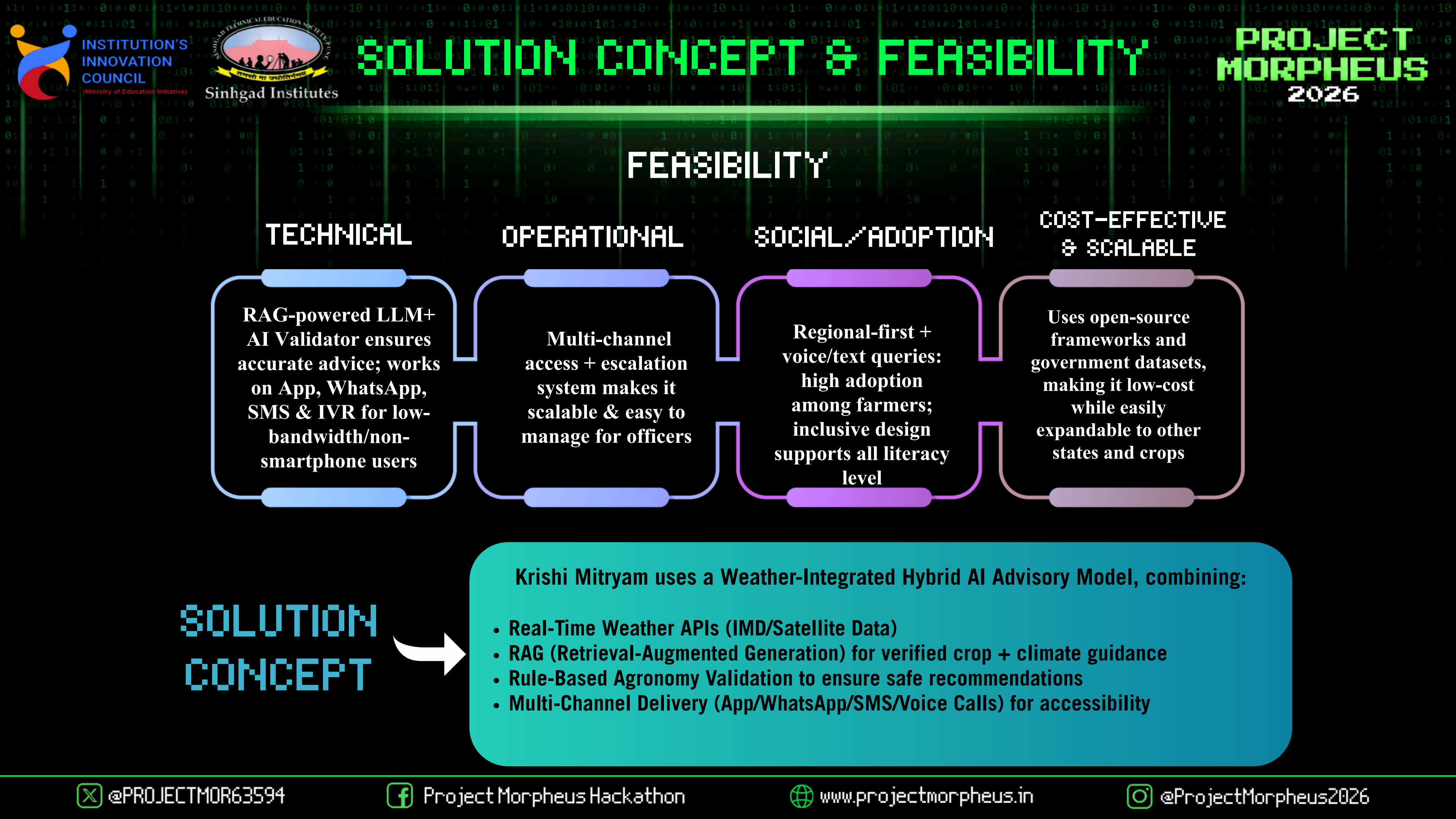
Offline-First & Low Connectivity Design

- Cached advisory packs for rural areas
- SMS fallback for no-internet zones
- Works even in remote villages.

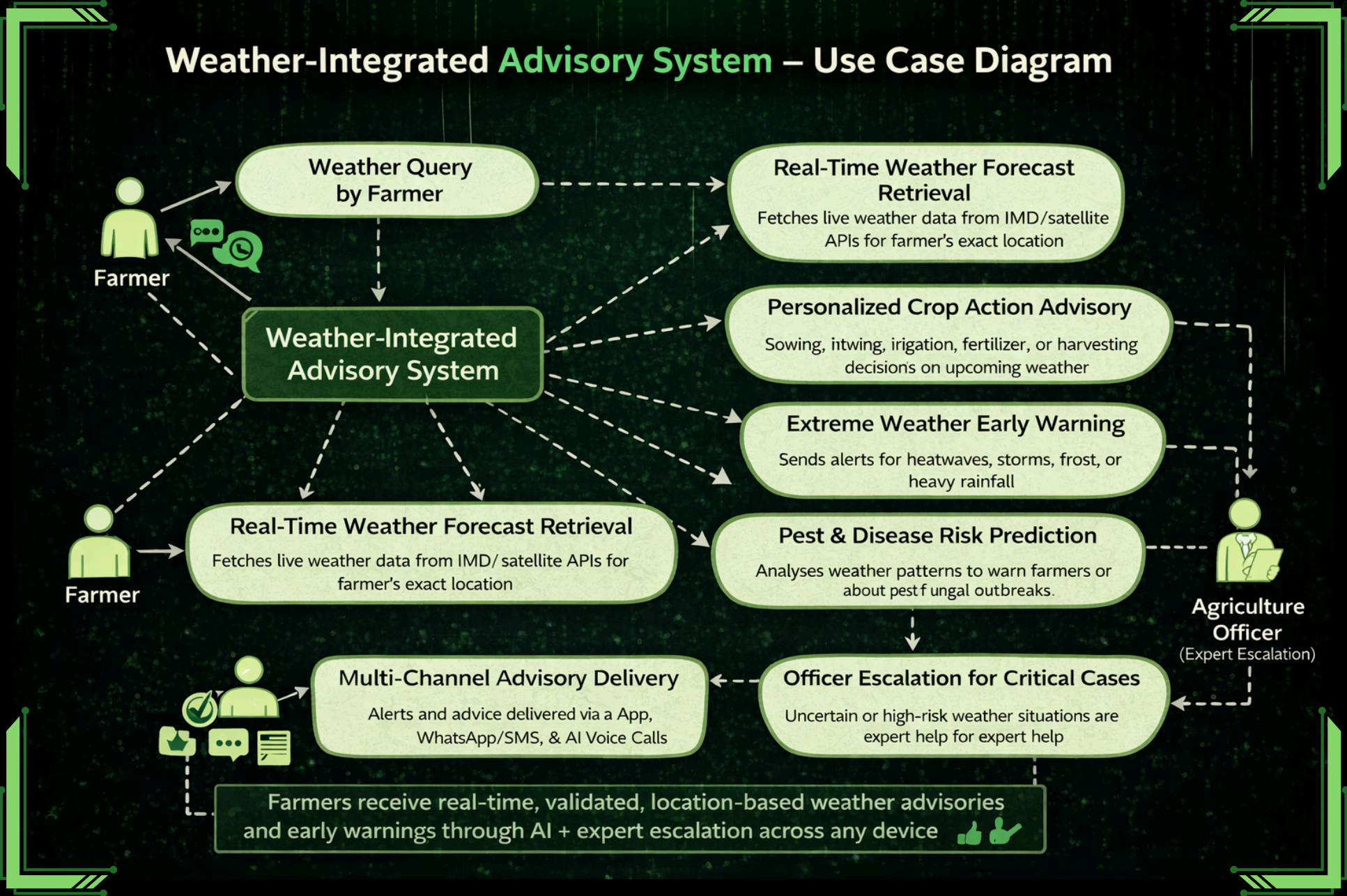
6

Escalation + Feedback Learning Loop

- Complex cases routed to human experts
- Farmer feedback improves model continuously
- AI + Human collaboration ensures trust.



1. Weather Query by Farmer
Farmers ask in local language (voice/text/WhatsApp) about rain, temperature, or crop safety.
2. Real-Time Weather Forecast Retrieval
System fetches live weather data from IMD/satellite APIs for the farmer's exact location.
3. Personalized Crop Action Advisory
AI suggests sowing, irrigation, fertilizer, or harvesting decisions based on upcoming weather.
4. Extreme Weather Early Warning
System sends alerts for heatwaves, storms, frost, or heavy rainfall to prevent crop damage.
5. Pest & Disease Risk Prediction
Weather patterns are analyzed to warn farmers about possible pest or fungal outbreaks.
6. Safe Recommendation Validation
Rule-based agronomy engine verifies pesticide dosage and scientifically safe practices.
7. Officer Escalation for Critical Cases
Uncertain or high-risk weather situations are forwarded to agriculture officers for expert help.
8. Multi-Channel Advisory Delivery
Alerts and advice delivered via App, WhatsApp/SMS, and AI Voice Calls for keypad users.
9. Feedback-Based Improvement
Farmer/officer feedback helps continuously improve future weather advisories.



FRONTEND



BACKEND



DATABASE & HOSTING

